

Specifications

SIZE About 360 x 250ft (110 x 75m)

WEIGHT About 925,000lb (420,000kg)

SPEED 17,500mph (28,000kph)

EARTH ORBIT TIME 90 minutes

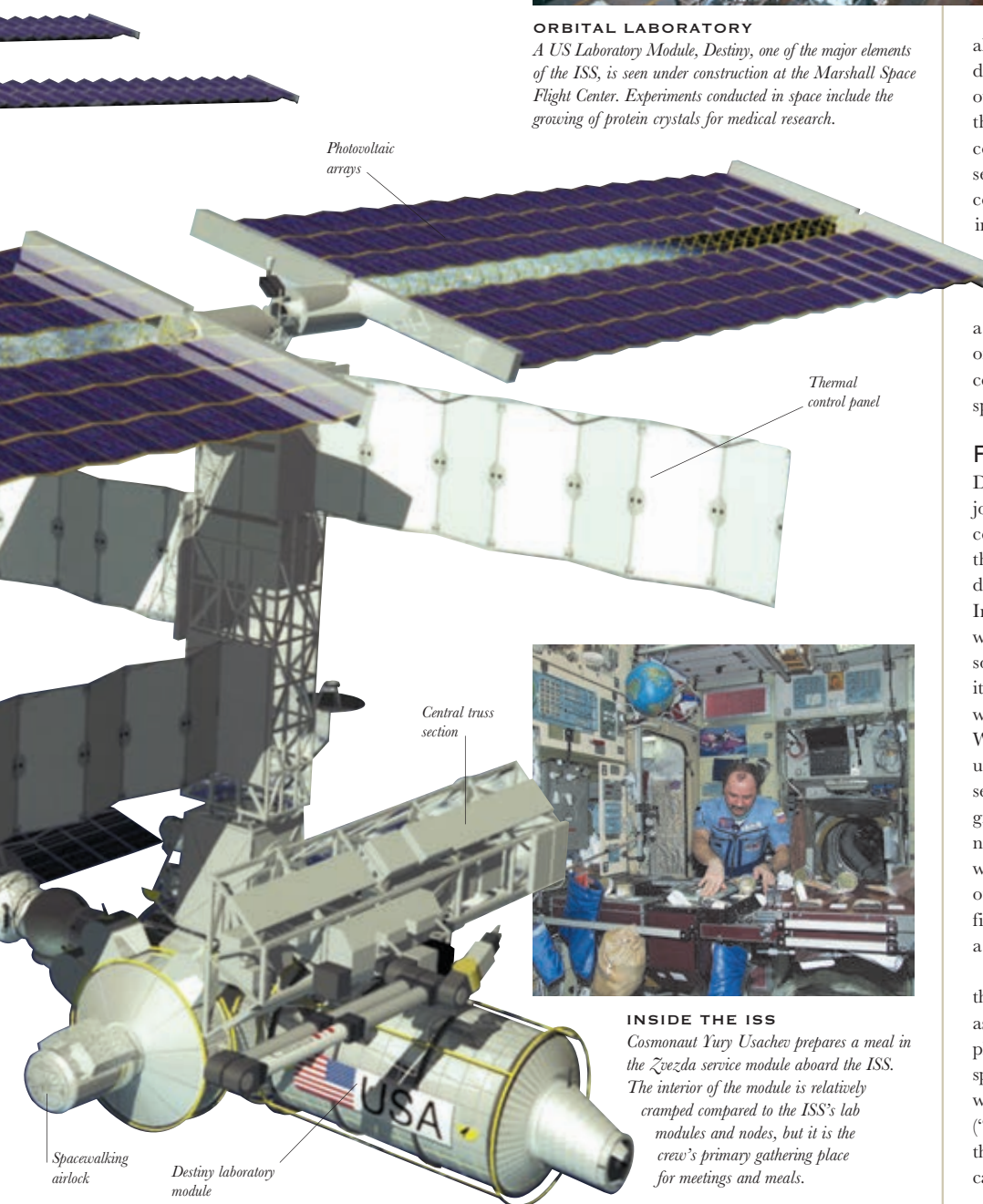
ALTITUDE 250 miles (410km)

CREW 6



ORBITAL LABORATORY

A US Laboratory Module, Destiny, one of the major elements of the ISS, is seen under construction at the Marshall Space Flight Center. Experiments conducted in space include the growing of protein crystals for medical research.



INSIDE THE ISS

Cosmonaut Yuri Usachev prepares a meal in the Zvezda service module aboard the ISS. The interior of the module is relatively cramped compared to the ISS's lab modules and nodes, but it is the crew's primary gathering place for meetings and meals.

The first American astronaut on board Mir was Norm Thagard, who stayed for 115 days from March to July 1995, and the last was Andy Thomas, from January to June 1998. It was, in the words of one NASA official, a critical exercise in "getting to know how to work with an international partner, getting to understand the Russian way of doing business." But it also brought warnings that Russian technology and operational expertise might not meet the standards that the Americans were used to.

By the time the joint American-Russian program began, Mir had been in orbit for almost a decade and much of its equipment was deteriorating, and in need of replacement. The overall quality of the Russian space program on the ground was also under threat from budgetary constraints. Mounting problems with Mir posed a serious risk to the lives of astronauts and cosmonauts. In June 1997 a cargo craft crashed into the space station, making a hole in its skin and taking out some solar panels. The following month the station lost power after one of the crew accidentally disconnected a cable. Other incidents in the same year included oxygen system failures, computer breakdowns, coolant leaks, and a fire on board that filled the space station with smoke.

Fiery end for Mir

Despite the dangers, NASA persisted with the joint Mir program until it was considered complete in the summer of 1998. The Americans then urged the Russians to abandon Mir and devote themselves singlemindedly to the incipient International Space Station project. But Mir had won a place in Russian hearts and was not to be so readily ditched. Cosmonauts continued to man it up to August 1999. Even after they left, there was a stay of execution when a consortium of Western businessmen put in a bid to buy Mir and use it as a space hotel. Whether this deal was a serious proposal for consideration, or not, Mir was given the death sentence. In March 2001 it was nudged lower by a Progress supply ship until it was close enough to the Earth to be pulled down out of orbit. Mir burned up in the atmosphere, fiery fragments crashing into the Pacific, ending a journey of over 2 billion miles (3 billion km).

By the time Mir hurtled from the sky, parts of the International Space Station had already been assembled in orbit and were providing a new platform for a permanent human presence in space. The project literally got off the ground when a Russian Proton rocket lifted the Zarya ("sunrise") module into orbit. The second section, the Unity node, linked up with Zarya a month later, carried into orbit by the Shuttle *Endeavour*. The